Mathematical Models of Infectious Diseases

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Contents

1 Logistics / Introduction **3**

2 Calculus **4**
2.1 Exponents and logarithms 4
2.1.1 Identities 4
2.2 The derivative as a linear approximation 4
2.2.1 Multiplication by a scalar 4
2.2.2 Other identities 5
2.3 The Fundamental Theorem of Calculus 5
2.3.1 Why this works 5

3 Probability **6**
3.1 Sample spaces and events 6
3.2 Set operations on events 6
3.3 The probability function 6
3.4 Probability of the complement 7
3.5 Conditional probabilities 7
3.6 Independence 7
3.7 Random variables 7
3.8 The exponential random variable 7
3.9 The memoryless property 8
3.10 Minimum of independent exponentials 8

4 The Basic SIR Model **9**
4.1 Compartments 9
4.2 Recovery: an exponential process 9
4.3 Adding new infections 10
4.4 Putting everything together 11

5 Linear Algebra **12**
5.1 Matrix-vector multiplication 12
5.2 The determinant as an area 12
5.3 Eigenvalues of a 2×2 matrix 13
5.4 Eigendecomposition 13
5.5 The determinant for a 3×3 matrix 13

6 Linear ODEs **14**
6.1 A single linear equation . . . 14
6.2 Why linearity matters . . . 14
6.3 Diagonal systems in two dimensions . . . 14
6.4 Diagonal systems with complex eigenvalues . . . 14
6.5 Every linear system is diagonal, in the right coordinates . . . 15

7 The SIR Model, Continued **16**
7.1 The threshold phenomenon . . . 16
7.1.1 The basic reproduction number . . . 16
7.2 Demographics . . . 16
7.2.1 R_0 for an SIR model with demographics . . . 17
7.3 Equilibria . . . 17
7.3.1 Equilibrium analysis for basic SIR model . . . 17
7.3.2 Equilibrium analysis for SIR model with demographics . . . 17
7.4 Stability analysis . . . 18
7.4.1 Linearization . . . 18
7.4.2 Linearization of SIR model . . . 19

8 Simulating Infectious Diseases **20**
8.1 Euler's method . . . 20
8.2 Incidence vs. prevalence . . . 20
8.3 Log-transformations . . . 21
8.4 A driven epidemic model . . . 22
8.5 Units . . . 22

9 State-Space Reconstruction **23**
9.1 States and Dynamics . . . 23
9.1.1 Prediction . . . 23
9.2 Takens' theorem . . . 24
9.2.1 Why $E > 2d$? (dimension counting) . . . 24
9.2.2 The Olsen–Schaffer attractor . . . 24
9.2.3 Prediction, revisited . . . 24

10 Convergent Cross Mapping **27**
10.1 Skew products . . . 27
10.2 Causality . . . 27
10.3 Convergent Cross Mapping . . . 27

11 Attractor Dimension **28**
11.1 What is a dimension . . . 28
11.2 Correlation dimension . . . 28
11.3 Levina–Bickel MLE estimator . . . 29

Chapter 1

Logistics / Introduction

Time. Mondays at 12:30pm.

Place. 209

Evaluation. Final exam 50%; Presentation 50%. You must pass the final exam to pass the course.

Preliminaries. Stundents are expected to be familiar with linear algebra, calculus and probability at a basic level. A refresher on these subjects will be given. Some promgramming will also be required and students are encouraged to employ AI models for programming.

Resources. The material pertaining to ODE based models is mostly based on *Modeling Infectious Diseases in Humans and Animals* by Matt Keeling and Pejman Rohani. The parts pertaining to State-Space Reconstruction methods cannot be find in books.

AI usage. Highly encouraged.

These notes are deliberately terse. Expect them to be clearer after you attend the discussions in the clasroom. These notes are were not proofread, so expect errors.

Chapter 2

Calculus

2.1 Exponents and logarithms

The exponential function is

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots$$

The function log is the inverse of the exponential:

$$\log e^x = x$$

For small x, truncating after the linear term gives the useful approximation

$$e^x \approx 1 + x \qquad (x \text{ small}).$$

Since log is the inverse of e^x, the same approximation, read the other way, gives

$$\log(1 + x) \approx x \qquad (x \text{ small}).$$

2.1.1 Identities

$$\begin{aligned}(e^x)^y &= e^{xy} \\ \log x^y &= y \log x \\ e^{x+y} &= e^x e^y.\end{aligned}$$

2.2 The derivative as a linear approximation

$$y + \Delta y = y(x + \Delta x) \approx y(x) + y'(x)\Delta x \quad \Longrightarrow \quad \Delta y = y'(x)\Delta x.$$

$y'(x)$ is called *the derivative*; it is also written $\dfrac{dy}{dx}$.

2.2.1 Multiplication by a scalar

What is the derivative of $z(x) = y(\alpha x)$? By the linear approximation:

$$z + \Delta z \approx z(x) + z'(x)\Delta x$$

But, on the other hand, by the definition of z:

$$\begin{aligned} z + \Delta z &= z(x + \Delta x) \\ &= y(\alpha x + \alpha \Delta x) \\ &= y(\alpha x) + y'(\alpha x)\alpha \Delta x \\ &= z(x) + y'(\alpha x) \cdot \alpha \Delta x \end{aligned}$$

Therefore

$$z'(x) = \alpha y'(\alpha x)$$

We will see later what this means for change of units. In the meantime, note a specific example: $(e^{\alpha x})' = \alpha e^{\alpha x}$.

2.2.2 Other identities

Sum rule:	$(y + z)'(x) = y'(x) + z'(x)$
Product / Leibniz rule:	$(yz)'(x) = y'(x)z(x) + y(x)z'(x)$
Chain rule:	$(y(z(x)))' = y'(z(x)) \cdot z'(x)$

2.3 The Fundamental Theorem of Calculus

$$\int_a^b y'(x)\,dx = y(b) - y(a).$$

2.3.1 Why this works

Partition $[a, b]$ into n steps $a = x_0 < x_1 < \cdots < x_n = b$, each of width Δx. By the linear approximation of Section 2, each step changes y by $\Delta y_i = y(x_{i+1}) - y(x_i) \approx y'(x_i)\Delta x$. Summing the steps telescopes:

$$\begin{aligned} \sum_{i=0}^{n-1} y'(x_i)\,\Delta x &\approx \sum_{i=0}^{n-1} \Delta y_i \\ &= \sum_{i=0}^{n-1} \underbrace{\big(y(x_{i+1}) - y(x_i)\big)}_{\Delta y_i} \\ &= \big(y(x_1) - y(x_0)\big) + \big(y(x_2) - y(x_1)\big) + \cdots + \big(y(x_n) - y(x_{n-1})\big) \\ &= y(x_n) - y(x_0) \\ &= y(b) - y(a). \end{aligned}$$

Example 1. *Since* $\frac{d}{ds}\left(-\frac{1}{\lambda}e^{-\lambda s}\right) = e^{-\lambda s}$, *for* $\lambda > 0$ *(so that* $e^{-\lambda s} \to 0$ *as* $s \to \infty$*),*

$$\int_t^\infty e^{-\lambda s}\,ds = \left[-\frac{1}{\lambda}e^{-\lambda s}\right]_t^\infty = \left(-\frac{1}{\lambda}e^{-\lambda \cdot \infty} + \frac{1}{\lambda}e^{-\lambda t}\right) = \frac{1}{\lambda}e^{-\lambda t}.$$

Chapter 3

Probability

3.1 Sample spaces and events

Definition 1. *An* experiment *is a process with uncertain outcome. The* sample space Ω *is the set of all possible outcomes. An* event *is a subset* $A \subseteq \Omega$.

Example 2. *A susceptible individual has one contact with an infectious person.*

$$\Omega = \{\textit{infected},\ \textit{not infected}\}.$$

$A = \{\textit{infected}\}$ *is an event.*

3.2 Set operations on events

- **Union** $A \cup B$: "A or B occurs."
- **Intersection** $A \cap B$: "A and B occur."
- **Complement** A^c: "A does not occur."
- **Disjoint**: $A \cap B = \emptyset$ (A and B cannot occur together).

Example 3. $A = \{\textit{infected with strain 1}\}$, $B = \{\textit{infected with strain 2}\}$.
- $A \cup B$*: infected with at least one strain.*
- $A \cap B$*: co-infected with both strains.*
- A^c*: not infected with strain 1.*

3.3 The probability function

Definition 2 (Axioms of probability)**.** Pr *assigns to each event* $A \subseteq \Omega$ *a number* $\Pr(A) \in [0, 1]$ *such that*
- $\Pr(\Omega) = 1$*;*
- *for disjoint events* A, B, $\Pr(A \cup B) = \Pr(A) + \Pr(B)$.

Interpretation: $\Pr(A)$ is the long-run fraction of times A occurs (e.g. $\Pr(\text{infection}) = 0.3$ means 30% of exposed contacts result in infection).

3.4 Probability of the complement

A very useful identity is $\Pr(A^c) = 1 - \Pr(A)$.

3.5 Conditional probabilities

$$\Pr(A|B) := \frac{\Pr(A \cap B)}{\Pr(B)} \qquad \text{whenever } \Pr(B) > 0.$$

3.6 Independence

Definition 3. *Events A and B are* independent *if*

$$\Pr(A \cap B) = \Pr(A)\ \Pr(B).$$

If moreover $\Pr(B) > 0$, this is equivalent (by the definition of conditional probability above) to $\Pr(A|B) = \Pr(A)$.

Interpretation: knowing that some of the events occurred gives no information about whether the others occurred.

Example 4. *A susceptible individual has n contacts. Each contact independently transmits infection with probability c. Let $A_i = \{$contact i transmits infection$\}$, so $\Pr(A_i) = q$. Avoiding infection on all n contacts is the event $A_1^c \cap \cdots \cap A_n^c$, so by independence*

$$\Pr(\textit{avoid infection}) = \prod_{i=1}^{n} \Pr(A_i^c) = (1-q)^n.$$

3.7 Random variables

Definition 4. *A* random variable *X is a function $X : \Omega \to \mathbb{R}$ assigning a number to each outcome.*

Example 5. *T = time until recovery.*

Definition 5. *The* density *(pdf) of X is a function $f \geq 0$ with $\int_{-\infty}^{\infty} f(x)\,dx = 1$ such that*

$$\Pr(a \leq X \leq b) = \int_a^b f(x)\,dx.$$

The expectation *is $E[X] = \int x f(x)\,dx$.*

3.8 The exponential random variable

Used to model waiting times: time to recovery, time until next contact, incubation period.

Definition 6. *T has an* exponential distribution with rate *$\lambda > 0$, written $T \sim Exp(\lambda)$, if*

$$f(t) = \lambda e^{-\lambda t}, \qquad t \geq 0.$$

The *survival function* is the probability of not yet having recovered by time t:

$$S(t) = \Pr(T > t) = \int_t^\infty \lambda e^{-\lambda s}\, ds = e^{-\lambda t}.$$

Mean: $E[T] = 1/\lambda$.

Interpretation: if T is the infectious period and $T \sim \text{Exp}(\gamma)$, then γ is the recovery *rate* and $1/\gamma$ is the mean infectious period.

3.9 The memoryless property

Proposition 1. *If $T \sim Exp(\lambda)$, then T is memoryless, i.e.*

$$\Pr(T > t + s \mid T > t) = \Pr(T > s) \qquad \textit{for all } s, t \geq 0.$$

Interpretation: given that an individual is still infectious at time t, the probability they remain infectious for s more days does not depend on t. Time already spent infected gives no information about time remaining — the recovery rate is constant, with no "aging" or "wearing out."

Proof.

$$\begin{aligned}
\Pr(T > t + s \mid T > t) &= \frac{\Pr(T > t + s \cap T > t)}{\Pr(T > t)} \\
&= \frac{\Pr(T > t + s)}{\Pr(T > t)} \\
&= \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} \\
&= e^{-\lambda s} \\
&= \Pr(T > s).
\end{aligned}$$

□

3.10 Minimum of independent exponentials

Proposition 2. *If $X \sim Exp(\lambda)$ and $Y \sim Exp(\eta)$ are independent, define $Z = \min\{X, Y\}$.*

$$Z \sim Exp(\lambda + \eta).$$

Interpretation: if an infected individual leaves the infected class either by recovering (rate γ) or by dying of unrelated causes (rate μ), independently, then the time spent in the infected class is $\min(T_{\text{recovery}}, T_{\text{death}}) \sim \text{Exp}(\gamma + \mu)$ — whichever happens first — with mean $1/(\gamma + \mu)$.

Proof. By independence

$$\begin{aligned}
\Pr(Z > t) &= \Pr(X > t)\ \Pr(Y > t) \\
&= e^{-\lambda t} e^{-\eta t} \\
&= e^{-(\lambda+\eta)t}
\end{aligned}$$

which is indeed the survival function of $\text{Exp}(\lambda + \eta)$. □

Chapter 4

The Basic SIR Model

4.1 Compartments

We assume infection is *acute*: an individual is infected, then is removed (recovers or dies) and stays removed. Three compartments:

- S — Susceptible: can become infected.
- I — Infected: currently infectious.
- R — Removed: no longer infectious, not reinfected.

We write X, Y, Z for the *number* of individuals in each class (this choice of notation will become clear later). The total population size N is assumed constant.

4.2 Recovery: an exponential process

Assume individual recovery happens at a constant rate γ, and for now ignore new infections. By Basic Probability, a constant recovery rate means each individual's remaining infectious time T is exponential: $T \sim \text{Exp}(\gamma)$.

Let $Y_0 := Y(0)$. The probability that an individual is still infected (not yet recovered) at time t is

$$P(T > t) = \int_t^\infty \gamma e^{-\gamma s}\, ds = e^{-\gamma t}.$$

Since each of the Y_0 initial infecteds recovers independently with this same law,

$$Y_t = Y_0\, e^{-\gamma t}.$$

Differentiating,

$$\dot{Y}_t = \frac{d}{dt}\left(Y_0 e^{-\gamma t}\right) = -\gamma Y_0 e^{-\gamma t} = -\gamma Y_t.$$

So with recovery alone, $\dot{Y} = -\gamma Y$.

What does γ mean? γ is the per-capita recovery rate. From probability, the infectious period is $T \sim \text{Exp}(\gamma)$, so the mean infectious period is $1/\gamma$. Example: if infected individuals recover after 5 days on average, $\gamma = 1/5$ day^{-1}.

4.3 Adding new infections

New infections add to Y; recoveries remove from Y:

$$\dot{Y} = (\text{rate of new infections}) - \underbrace{\gamma Y}_{\text{recovery rate}}.$$

Let us assume for now that there are no recoveries. We derive the rate of new infections from first principles. Let κ denote the number of contacts per time unit for a susceptible individual. So during a short time interval this individual has $\kappa\frac{Y}{N} \times \delta t$ contacts with infecteds. We let

$$c = \Pr(\text{transmission}|\text{contact with infected}).$$

Hence $1 - c$ is the complement (no transmission given a contact) and

$$\Pr(\text{no infection during time } \delta t) = (1-c)^{\kappa\frac{Y}{N}\delta t}$$

and so

$$\begin{aligned}
\Pr(\text{infection during time } \delta t) &= 1 - (1-c)^{\kappa\frac{Y}{N}\delta t} \\
&= 1 - e^{-\beta\frac{Y}{N}\delta t} \quad (\text{let } \beta := -\kappa\log(1-c)) \\
&= 1 - (1 - \beta\frac{Y}{N}\delta t) \quad (\text{since } e^x \approx 1 + x \text{ for small x}) \\
&= \beta\frac{Y}{N}\delta t
\end{aligned}$$

Definition 7. *The* force of infection *(FOI) λ_t is the per-capita rate at which susceptibles become infected.*

It is clear that $\lambda_t = \beta\frac{Y}{N}$. This specific form is called a mass-action force of infection. We can see this by multiplying by the number of susceptibles X:

$$\begin{aligned}
\text{number of new infections during } \delta t &= X \Pr(\text{infection during time } \delta t) \\
&= \beta\frac{Y}{N}\delta t \cdot X
\end{aligned}$$

so the rate of new infections is

$$\text{rate of new infections} = \frac{\beta\frac{Y}{N}\delta t}{\delta t}X = \beta\frac{Y}{N}X.$$

What does β mean? See below.

What assumptions did we make? Contacts do not depend on the infection state of individuals; the number of contacts between individuals does not depend on population size N: doubling both N and Y leaves λ_t unchanged.

4.4 Putting everything together

We dropped recoveries above only to isolate the derivation of the infection rate; reinstating the recovery term γY from Section 2, we have

$$\begin{aligned}
\dot{X} &= -\text{rate of new infections} = -\beta \frac{Y}{N} X, \\
\dot{Y} &= \text{rate of new infections} - \text{rate of recovery} = \beta \frac{Y}{N} X - \gamma Y, \\
\dot{Z} &= \text{rate of recovery} = \gamma Y.
\end{aligned}$$

Dividing each equation by N (constant), and writing $S = X/N$, $I = Y/N$, $R = Z/N$ for the population proportions gives the standard formulation of the SIR model:

$$\begin{aligned}
\dot{S} &= -\beta SI, \\
\dot{I} &= \beta SI - \gamma I, \\
\dot{R} &= \gamma I.
\end{aligned}$$

Note $\dot{S} + \dot{I} + \dot{R} = 0$: since S, I, R are proportions, $S + I + R = 1$ is constant. Hence one equation is redundant — given any two, the third follows. It is common to drop the $\dot{R}$ equation and recover $R = 1 - S - I$ when needed.

What does β mean? If $S_0 \approx 1$ and $I_0 > 0$ then in one time unit we will have $\beta SI \approx \beta I$ new infections. So *per individual* we will have β new infections. Thus, β is the number of new infections per individual per time unit that an infected individual can infect.

Chapter 5

Linear Algebra

5.1 Matrix-vector multiplication

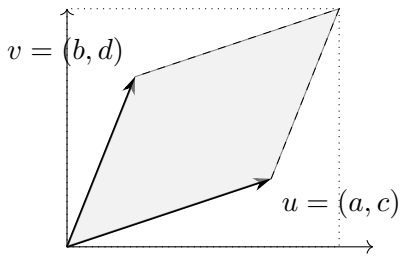
A 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ acts on a vector $v = \begin{pmatrix} x \\ y \end{pmatrix}$ by

$$Av = x \begin{pmatrix} a \\ c \end{pmatrix} + y \begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

That is, Av is the combination of A's two columns, $\begin{pmatrix} a \\ c \end{pmatrix}$ and $\begin{pmatrix} b \\ d \end{pmatrix}$, weighted by v's entries x, y.

5.2 The determinant as an area

Take two vectors in the plane, $u = (a, c)$ and $v = (b, d)$ — the columns of A from Section 1. They span a parallelogram.



Enclose the parallelogram in the axis-aligned bounding box of width $a+b$ and height $c+d$. The box consists of the parallelogram, two right triangles with legs a, c (combined area ac), two right triangles with legs b, d (combined area bd), and two $b \times c$ rectangles (combined area $2bc$). So

$$(a + b)(c + d) = \text{Area} + ac + bd + 2bc \quad \Longrightarrow \quad \text{Area} = ad - bc.$$

Definition 8. *The* determinant *of a matrix* $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ *is the* signed *area spanned by* $u = (a, c)$, $v = (b, d)$.

We had just saw that $\det A = ad - bc$. Geometrically: $\det A = 0$ exactly when the columns of A are parallel — the parallelogram they span is flat, with zero area.

5.3 Eigenvalues of a 2×2 matrix

Definition 9. *A nonzero vector v is an* eigenvector *of A with* eigenvalue λ *if $Av = \lambda v$.*

Rearranging, $(A - \lambda I)v = 0$ for some $v \neq 0$. We saw that $(A - \lambda I)v$ is the combination of $(A - \lambda I)$'s two columns weighted by v's entries; for a *nonzero* v to give 0, that combination must be nontrivial, which is only possible if the two columns are parallel, i.e.

$$\det(A - \lambda I) = 0.$$

Useful fact. The determinant is the product of eigenvalues, whereas the trace is their sum.

5.4 Eigendecomposition

Suppose A has eigenvectors v_1, v_2 with eigenvalues λ_1, λ_2, and let $P = \begin{pmatrix} v_1 & v_2 \end{pmatrix}$ (eigenvectors as columns). Then

$$AP = \begin{pmatrix} Av_1 & Av_2 \end{pmatrix} = \begin{pmatrix} \lambda_1 v_1 & \lambda_2 v_2 \end{pmatrix} = PD, \qquad D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix},$$

so, if v_1, v_2 are independent (P invertible),

$$A = PDP^{-1}.$$

P is a *change of basis*: P^{-1} converts a vector from the standard basis to *eigenbasis* coordinates (coefficients along v_1, v_2), and P converts back. In eigenbasis coordinates A acts as the diagonal D.

5.5 The determinant for a 3×3 matrix

$$\det \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} = a_1 \det \begin{pmatrix} b_2 & c_2 \\ b_3 & c_3 \end{pmatrix} - a_2 \det \begin{pmatrix} b_1 & c_1 \\ b_3 & c_3 \end{pmatrix} + a_3 \det \begin{pmatrix} b_1 & c_1 \\ b_2 & c_2 \end{pmatrix}.$$

The same interpretations hold: the determinant is (signed) volume, the determinant is the product of eigenvalues, and the trace is the sum of eigenvalues.

Chapter 6

Linear ODEs

6.1 A single linear equation

$$\dot{x} = ax, \qquad x(0) = x_0.$$

Solve: $\frac{d}{dt}\log x = \dot{x}/x = a$, so $\log x = \log x_0 + at$, i.e.

$$x(t) = x_0 e^{at}.$$

$a > 0$: exponential growth. $a < 0$: exponential decay. $a = 0$: constant.

6.2 Why linearity matters

A system $\dot{x} = Ax$ ($x \in \mathbb{R}^n$, A a matrix) is *linear* in x. Its key property: if $x_1(t), x_2(t)$ both solve $\dot{x} = Ax$, so does any combination $c_1x_1 + c_2x_2$, since

$$\frac{d}{dt}(c_1x_1 + c_2x_2) = c_1\dot{x}_1 + c_2\dot{x}_2 = c_1Ax_1 + c_2Ax_2 = A(c_1x_1 + c_2x_2).$$

6.3 Diagonal systems in two dimensions

A *diagonal* system decouples into two copies of Section 1:

$$\dot{x} = \lambda_1 x, \qquad \dot{y} = \lambda_2 y \quad \Longrightarrow \quad x(t) = x_0 e^{\lambda_1 t}, \;\; y(t) = y_0 e^{\lambda_2 t}.$$

The qualitative picture depends on the signs of λ_1, λ_2:

6.4 Diagonal systems with complex eigenvalues

Take $A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, i.e. $\dot{x} = ax - by$, $\dot{y} = bx + ay$. Its eigenvalues solve $\det(A - \lambda I) = (a-\lambda)^2 + b^2 = 0$, so $\lambda = a \pm ib$: complex. Set $z = x + iy$; then

$$\dot{z} = \dot{x} + i\dot{y} = (ax - by) + i(bx + ay) = (a + ib)(x + iy) = (a + ib)z,$$

the single linear equation of Section 1, now with complex $a + ib$. Its solution is $z(t) = z_0 e^{(a+ib)t} = z_0 e^{at}\big(\cos bt + i \sin bt\big)$: a rotation at rate b, scaled by e^{at}.

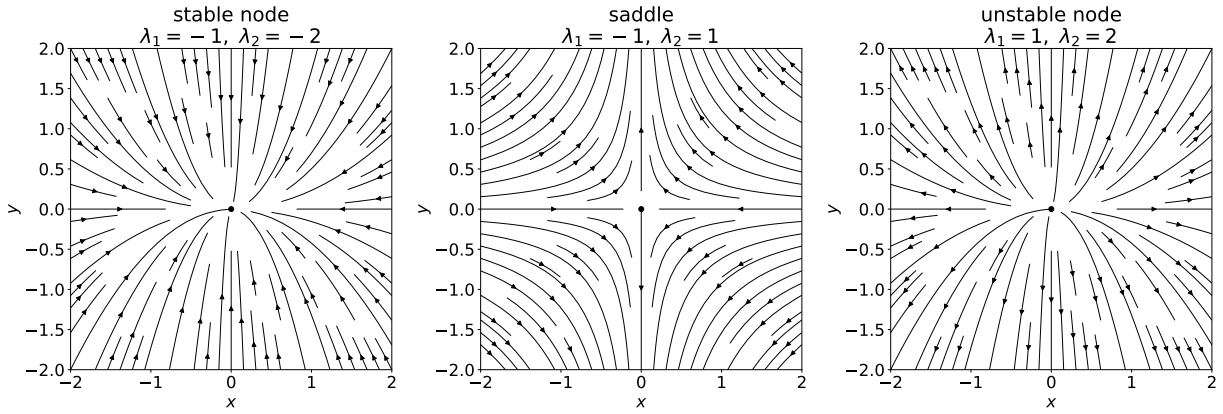


Figure 6.1: Both negative: every trajectory decays to the origin (*stable node*). Both positive: every trajectory grows away from it (*unstable node*). Opposite signs: the origin attracts along one axis and repels along the other (*saddle*).

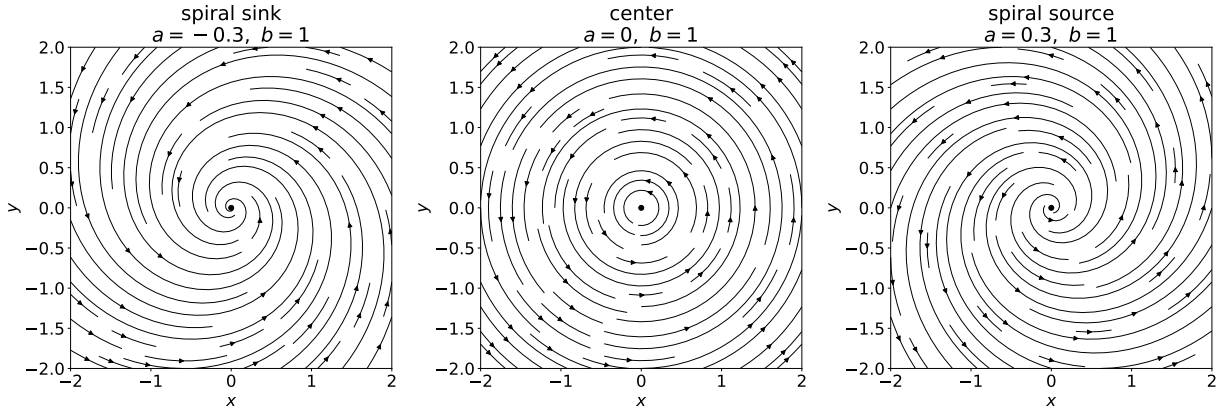


Figure 6.2: $a < 0$: spiral sink. $a = 0$: pure rotation, a *center*. $a > 0$: spiral source.

6.5 Every linear system is diagonal, in the right coordinates

Let A have eigenvectors v_1, v_2 (eigenvalues λ_1, λ_2, possibly complex conjugates) and $P = \begin{pmatrix} v_1 & v_2 \end{pmatrix}$ as in the eigendecomposition $A = PDP^{-1}$. Write $x = Pu$, i.e. $u = P^{-1}x$ gives the coordinates of x in the eigenbasis. Then

$$\dot{u} = P^{-1}\dot{x} = P^{-1}Ax = P^{-1}APu = Du.$$

So u solves a *diagonal* system — Section 3 if λ_1, λ_2 are real, Section 4 if they are complex conjugates — and $x(t) = Pu(t)$ translates the solution back to the original coordinates. Every linear system is, up to a change of basis, one of the pictures above.

Chapter 7

The SIR Model, Continued

7.1 The threshold phenomenon

Say the initial proportion of infecteds is $I_0 \neq 0$. Will the infection *invade*? I.e. will I increase? Equivalently, does $\dot{I} > 0$ hold?

$$\dot{I} = \beta SI - \gamma I = I(\beta S - \gamma).$$

So $\dot{I} > 0$ if

$$\beta S - \gamma > 0.$$

We must have $\beta S - \gamma > 0$, or equivalently

$$S > \frac{\gamma}{\beta}.$$

So if $S < \gamma/\beta$ the infection cannot invade. The implication for vaccination strategies and herd immunity is immediate.

7.1.1 The basic reproduction number

What is β/γ? Recall $1/\gamma$ is the average time to recovery, i.e. the disease duration and β is infections per individual per time unit (in a fully susceptible population). Hence R_0 is the expected number of secondary infections caused by a single infected individual introduced into an otherwise fully susceptible population. We denote it

$$R_0 = \frac{\beta}{\gamma}$$

7.2 Demographics

We want to include births and deaths. Individuals are born into the susceptible class S at rate μ (per unit of the whole population) and die at the same per-capita rate μ from every class, so population size is still fixed. The new equations are

$$\begin{aligned}
\dot{S} &= \mu - \beta SI - \mu S, \\
\dot{I} &= \beta SI - \gamma I - \mu I, \\
\dot{R} &= \gamma I - \mu R.
\end{aligned}$$

As before, $\dot{S} + \dot{I} + \dot{R} = 0$, so $S + I + R = 1$ stays constant.

7.2.1 R_0 for an SIR model with demographics

With demography, an infected individual leaves the infected class either by recovering (rate γ) or by dying (rate μ), independently. Recall that *minimum* of two independent exponentials is exponential with the sum of rates, so the time spent infected is $\text{Exp}(\gamma + \mu)$.

Using the same interpretation as before, R_0 is (contacts during infection) $\times$ (transmission probability per contact) with the mean infectious period $1/(\gamma + \mu)$:

$$R_0 = \frac{\beta}{\gamma + \mu}.$$

7.3 Equilibria

We find equilibrium states by equating the derivatives to zero.

7.3.1 Equilibrium analysis for basic SIR model

Recall the basic SIR model:

$$\begin{aligned}\dot{S} &= -\beta SI,\\ \dot{I} &= \beta SI - \gamma I,\\ \dot{R} &= \gamma I.\end{aligned}$$

Set all three derivatives to zero. $\dot{R} = \gamma I = 0$ forces $I = 0$ (since $\gamma > 0$). Then $\dot{S} = \dot{I} = 0$.

So the equilibria are $(S, 0, R)$ for $S + R = 1$ (and, of course, $0 \le S \le 1$). This is a *disease-free equilibrium* (DFE).

7.3.2 Equilibrium analysis for SIR model with demographics

Recall the model with demography:

$$\begin{aligned}\dot{S} &= \mu - \beta SI - \mu S,\\ \dot{I} &= \beta SI - \gamma I - \mu I,\\ \dot{R} &= \gamma I - \mu R.\end{aligned}$$

Set all three derivatives to zero. From $\dot{I} = 0$: $I[\beta S - (\gamma + \mu)] = 0$, so either $I = 0$ or $S = \dfrac{\gamma + \mu}{\beta} = \dfrac{1}{R_0}$.

Disease-free equilibrium (DFE). Then $\dot{S} = \mu(1 - S) = 0$ forces $S = 1$. Other compartments are empty.

Endemic equilibrium (EE). Take $\dot{I} = 0$. If $I = 0$ we are at DFE. So assume $I > 0$. Then

$$\dot{I} = 0 \Rightarrow \beta S^* = \gamma + \mu \Rightarrow S^* = \frac{\gamma + \mu}{\beta} = \frac{1}{R_0}.$$

Then

$$I^* = \frac{\mu}{\gamma + \mu}\left(1 - \frac{1}{R_0}\right)$$

and

$$R^* = \frac{\gamma}{\gamma+\mu}\left(1 - \frac{1}{R_0}\right).$$

So

$$(S^*, I^*, R^*) = \left(\frac{1}{R_0},\ \ \frac{\mu}{\gamma+\mu}\Big(1-\frac{1}{R_0}\Big),\ \ \frac{\gamma}{\gamma+\mu}\Big(1-\frac{1}{R_0}\Big)\right).$$

Note that this equilibrium is feasible (i.e. $I^* > 0$) exactly when $R_0 > 1$.

7.4 Stability analysis

7.4.1 Linearization

For a real function of one variable, near a point x^*:

$$f(x^* + \delta x) \approx f(x^*) + \frac{df}{dx}(x^*)\,\delta x.$$

In two dimensions a function takes two inputs and produces two outputs $(x, y) \mapsto (f(x, y), g(x, y))$. So:

$$f(x^* + \delta x,\, y^* + \delta y) \approx f(x^*, y^*) + \frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial y}\delta y$$
$$g(x^* + \delta x,\, y^* + \delta y) \approx g(x^*, y^*) + \frac{\partial g}{\partial x}\delta x + \frac{\partial g}{\partial y}\delta y.$$

Collecting:

$$\begin{pmatrix} f(x^*+\delta x, y^*+\delta y) \\ g(x^*+\delta x, y^*+\delta y)\end{pmatrix} \approx \begin{pmatrix} f(x^*, y^*) \\ g(x^*, y^*)\end{pmatrix} + \begin{pmatrix} \partial f/\partial x & \partial f/\partial y \\ \partial g/\partial x & \partial g/\partial y\end{pmatrix}\begin{pmatrix}\delta x \\ \delta y\end{pmatrix}.$$

Definition 10. *The* Jacobian *of* (f, g) *at* (x^*, y^*) *is*

$$J(x^*, y^*) = \begin{pmatrix} \partial f/\partial x & \partial f/\partial y \\ \partial g/\partial x & \partial g/\partial y\end{pmatrix},$$

So:

$$\begin{pmatrix} f(x^*+\delta x, y^*+\delta y) \\ g(x^*+\delta x, y^*+\delta y)\end{pmatrix} \approx \begin{pmatrix} f(x^*, y^*) \\ g(x^*, y^*)\end{pmatrix} + J(x^*, y^*)\begin{pmatrix}\delta x \\ \delta y\end{pmatrix}.$$

Now, our goal is to linearize a dynamical system

$$\begin{pmatrix}\dot{x} \\ \dot{y}\end{pmatrix} = \begin{pmatrix} f(x, y) \\ g(x, y)\end{pmatrix}$$

using the Jacobian. We linearize at an equilibrium point, $f(x^*, y^*) = g(x^*, y^*) = 0$ and let $x = x^* + \delta x$, $y = y^* + \delta y$. Then

$$\begin{pmatrix}\dot{x} \\ \dot{y}\end{pmatrix} = \begin{pmatrix} f(x^*, y^*) \\ g(x^*, y^*)\end{pmatrix}$$

Since x^*, y^* are constant, $\dot{x} = \dot{\delta x}$ and $\dot{y} = \dot{\delta y}$, so by the approximation above and using $f(x^*, y^*) = g(x^*, y^*) = 0$:

$$\begin{pmatrix} \dot{\delta x} \\ \dot{\delta y} \end{pmatrix} \approx J(x^*, y^*) \begin{pmatrix} \delta x \\ \delta y \end{pmatrix}.$$

Near an equilibrium, the dynamics of a small perturbation are therefore approximately *linear*, governed entirely by the Jacobian.

Why is this useful? Stability of an equilibrium is "stable" determined by its eigenvalues. Roughly, all eigenvalues have to be negative[1] for an equilibrium to be stable.

7.4.2 Linearization of SIR model

Here the variables are S and I:

$$\dot{S} = f(S, I) := \mu(1 - S) - \beta SI$$
$$\dot{I} = g(S, I) := \beta SI - (\gamma + \mu)I$$

The Jacobian is

$$J(S, I) = \begin{pmatrix} -\mu - \beta I & -\beta S \\ \beta I & \beta S - (\gamma + \mu) \end{pmatrix}.$$

Linearization around the disease-free equilibrium (DFE)

In the disease-free equilibrium $I^* = 0$ and $S^* = 1$:

$$J(S^*, I^*) = \begin{pmatrix} -\mu & -\beta \\ 0 & \beta - (\gamma + \mu) \end{pmatrix}$$

Eigenvalues are:

$$\lambda_1 = -\mu, \qquad \lambda_2 = \beta - (\gamma + \mu) = (\gamma + \mu)(R_0 - 1).$$

$\lambda_1 < 0$ always; $\lambda_2 < 0 \iff R_0 < 1$. So the disease-free equilibrium is stable iff $R_0 < 1$.

Linearization around the endemic equilibrium (EE)

In the endemic equilibrium,

$$(S^*, I^*, R^*) = \left(\frac{1}{R_0}, \ \frac{\mu}{\gamma + \mu}\Big(1 - \frac{1}{R_0}\Big), \ \frac{\gamma}{\gamma + \mu}\Big(1 - \frac{1}{R_0}\Big) \right).$$

$\beta S^* = \gamma + \mu$ and $\beta I^* = \mu(R_0 - 1)$ and the Jacobian becomes

$$J(S^*, I^*) = \begin{pmatrix} -\mu R_0 & -(\gamma + \mu) \\ \mu(R_0 - 1) & 0 \end{pmatrix}, \qquad \operatorname{tr} J(S^*, I^*) = -\mu R_0, \qquad \det J(S^*, I^*) = \mu(\gamma + \mu)(R_0 - 1).$$

For a 2×2 matrix, both eigenvalues have negative real part iff $\operatorname{tr} < 0$ and $\det > 0$. Here $\operatorname{tr} J^* < 0$ always, and $\det J^* > 0 \iff R_0 > 1$. So whenever the endemic equilibrium exists ($R_0 > 1$), it is automatically stable.

Summary. $R_0 < 1 \Rightarrow$ disease-free equilibrium is stable. $R_0 > 1 \Rightarrow$ endemic equilibrium is stable.

[1] have negative real part.

Chapter 8

Simulating Infectious Diseases

8.1 Euler's method

Recall $f(x + \Delta x) \approx f(x) + f'(x)\Delta x$. *Euler's method* uses this repeatedly for $\dot{x} = f(x, t)$:

$$x_{t+\Delta t} \approx x_t + f(x_t, t)\,\Delta t,$$

then repeat. For an SIR system, $x_t = (S, I, R)$:

```
import numpy as np

def euler(beta, gamma, dt, n):
    S, I = np.zeros(n), np.zeros(n)
    S[0], I[0] = 0.99, 0.01
    for t in range(n - 1):
        dS = - beta * S[t] * I[t]
        dI = beta * S[t] * I[t] - gamma * I[t]

        # Euler's method: new state = old state + rate * step size
        S[t + 1] = S[t] + dS * dt
        I[t + 1] = I[t] + dI * dt

    return S, I
```

8.2 Incidence vs. prevalence

I_t is the per-capita *prevalence*: the proportion infected at a time t. Surveillance rarely measures prevalence. Instead, surveillance efforts reoirt *incidence*: the number of *new* infections over a reporting interval, e.g. a week:

Definition 11. *The* incidence *over an interval* $[t - \Delta t,\, t]$ *is the number of new infections:*

$$\mathit{Incidence}(t) = N \int_{t-\Delta t}^{t} \lambda_s\, S_s\, ds.$$

Tyically we normalize to find the *per-capita* incidence

```
import numpy as np
```

```
def incidence_euler(beta, gamma, dt, k, n):
    S, I = 0.99, 0.01
    incidence = np.zeros(n)

    for t in range(n-1):
        for s in range(k):
            new_infections = beta * S * I
            dS = - new_infections
            dI = new_infections - gamma * I

            # Euler's method: new state = old state + rate * step size
            S += dS * dt
            I += dI * dt

            # Accumulate new infections
            incidence[t+1] += new_infections * dt

    return incidence
```

Note: since the Euler time step (e.g. day) may differ from the sampling rate[1] (e.g. one week) we had to implement another loop that rubs over $0, ..., k$.

8.3 Log-transformations

Working with log variables is more stable numeically. We can easily find that

$$\begin{aligned}
\dot{\log}(S) &= \frac{1}{S}\dot{S} \\
&= \frac{-\beta SI}{S} \\
&= -\beta e^{\log(I)} \\
\dot{\log}(I) &= \frac{\beta SI - \gamma I}{I} \\
&= -\beta e^{\log(S)} - \gamma.
\end{aligned}$$

```
import numpy as np

def log_euler(beta, gamma, dt, n):
    logS, logI = np.zeros(n), np.zeros(n)
    logS[0], logI[0] = np.log(0.99), np.log(0.01)
    for t in range(n - 1):
        dlogS = - beta * np.exp(logI[t])
        dlogI = beta * np.exp(logS[t]) - gamma

        # Euler's method: new state = old state + rate * step size
        logS[t + 1] = logS[t] + dlogS * dt
        logI[t + 1] = logI[t] + dlogI * dt

    return S, I
```

[1] To be precise, the Euler step may differ from the *inverse* of the sampling rate.

8.4 A driven epidemic model

Many infectious diseases are seasonal (e.g. measles and influenza), since transmission varies in time. We let β vary periodically in time:

$$\beta_t = \beta_0\big(1 + \beta_1 \cos(2\pi t)\big),$$

This is a *driven*, or *forced* system, with t in years, $\beta_0 > 0$ the baseline transmission rate, and $\beta_1 \in [0, 1)$ the amplitude of seasonal forcing. Substituting into the demographic SIR system:

$$\begin{aligned}
\dot{S} &= \mu(1 - S) - \beta_t\, SI, \\
\dot{I} &= \beta_t\, SI - (\gamma + \mu)I, \\
\dot{R} &= \gamma I - \mu R.
\end{aligned}$$

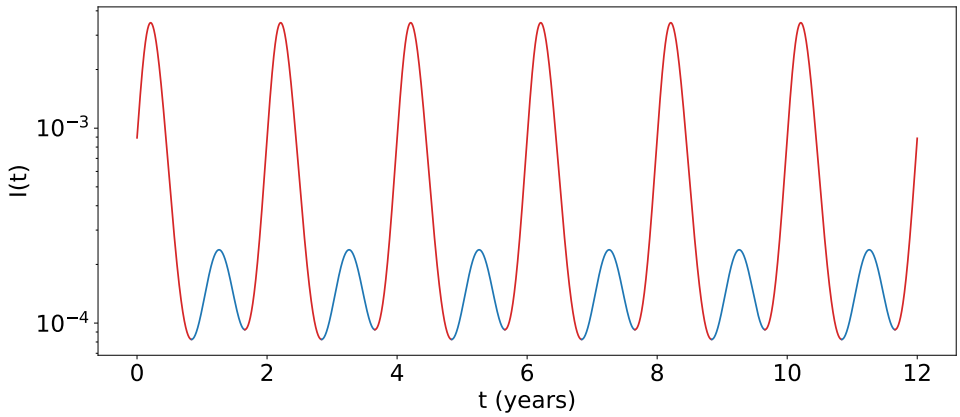


Figure 8.1: A biennial cycle: one large outbreak (red), one small (blue), repeating exactly every two years (log scale — the small outbreaks are $\sim 15\times$ smaller). Parameters: $\mu = 0.02\,\mathrm{yr}^{-1}$, $\gamma = 365/13\,\mathrm{yr}^{-1}$ (13-day infectious period), $R_0 = 17$ (so $\beta_0 = R_0(\gamma + \mu) \approx 477.6\,\mathrm{yr}^{-1}$), $\beta_1 = 0.185$.

8.5 Units

Units are crucial in simulations and modeling. We first decide on time unit (e.g. year, month, week) and then derive parameters accordingly. See Table 8.1 for examples.

description	value
recovery rate	$\gamma = \frac{1}{13}\,\mathrm{day}^{-1} = \frac{1}{13}\left(\frac{\mathrm{year}}{365}\right)^{-1} = \frac{365}{13}\,\mathrm{year}^{-1}$
birth rate	$\mu = \frac{1}{50}\,\mathrm{year}^{-1} = \frac{1}{50}(365\,\mathrm{day})^{-1} = \frac{1}{18250}\,\mathrm{day}^{-1}$
contact rate	$\kappa = 10\,\mathrm{day}^{-1} = 10\left(\frac{\mathrm{year}}{365}\right)^{-1} = 3650\,\mathrm{year}^{-1}$

Table 8.1: Converting parameter values between day and year units, by treating day and year as algebraic symbols (day = year/365).

Chapter 9

State-Space Reconstruction

9.1 States and Dynamics

Define $\phi_t := 2\pi t \bmod 2\pi$. If we know S, I and ϕ at one time point we fully know the future. The *state-space* of the system is

$$\Omega = (S, I, \phi) \in \mathbb{R}^2 \times \mathbb{S}^1.$$

To find $\omega_{t+1} = (S_{t+1}, I_{t+1}, \phi_{t+1})$ from $\omega_t = (S_t, I_t, \phi_t)$ we integrate the SIR equations. This *law* is encapsulated in an operator $T\omega_t = \omega_{t+1}$.

Definition 12. *A* dynamical system *is a pair* (Ω, T) *where* Ω *called the* state-space *and* $T : \Omega \to \Omega$ *is the* law *of the dynamics.*

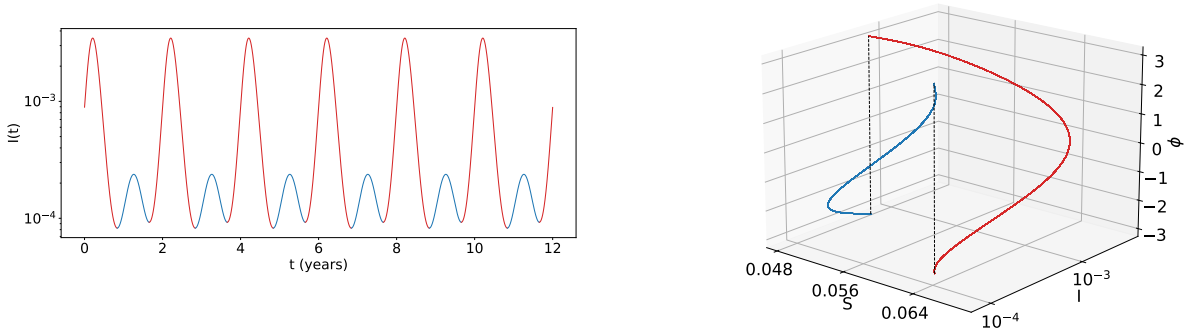


Figure 9.1: The trajectory of the driven SIR model in time (left) and in state-space (S, I, ϕ) (right). Same parameters as Figure 8.1, red = large outbreak year, blue = small).

9.1.1 Prediction

Say we know ω_t and we have many records of $\omega_s, s < t$. What we do not have is access to T itself. How would we predict ω_{t+1}? Easy: find $t^\star = \arg\min_{s<t} \|\omega_s - \omega_t\|$ and predict $\omega_{t+1} \approx \omega_{t^\star+1}$.

9.2 Takens' theorem

Definition 13. *An* observable *is a real-valued function $X : \Omega \to \mathbb{R}$. We usually abuse notation and write $X_t = X(\omega_t)$.*

In many applications we do not have access to neither T nor to Ω. However, we may have records of some observable(s). Let X_t be an observed quantity (e.g. fraction of infecteds I_t), fix a lag L and an *embedding dimension* E. The *delay vector* is

$$x_t = \big(X_t,\, X_{t-L},\, \ldots, X_{t-(E-1)L}\big).$$

It is easiest to take L to be the spacing between observations (i.e. inverse of the sampling rate).

Theorem 1 (Takens, 1981)**.** *Let (Ω, T) a dynamical system, $d = \dim \Omega$ and $X : \Omega \to \mathbb{R}$ an observable, $L > 0$ some lag, and $E > 2d$. Then, generically, the map $x_t \to \omega_t$ is a diffeomorphism (everywhere differentiable with an inverse which is also everywhere differentiable).*

In lignt of Takens' Theorem, we call x_t *reconstructed states*. We denote the set of all reconstruted states $M_X := \{x_s : s \le t\}$.

9.2.1 Why $E > 2d$? (dimension counting)

The map $x_t \to \omega_t$ can fail in two ways, and both are ruled out by a dimension count once $E > 2d$.

- **Injectivity.** $x_s = x_t$ for $s \neq t$ means E equations $X(\omega_{s-kL}) = X(\omega_{t-kL})$, $k = 0, \ldots, E-1$, hold at once. These are imposed on the $2d$-dimensional space of pairs $(\omega_s, \omega_t) \in \Omega \times \Omega$. With $E > 2d$ this is *overdetermined* — generically, more equations than degrees of freedom means no solutions.
- **Immersion.** The map fails to be locally invertible at ω_t exactly when some direction $v \in T_{\omega_t}\Omega$ is killed by all E copies of dX pulled back along the orbit. Again, generically, requiring E independent conditions to vanish on a space of dimension $d-1$ forces $v = 0$ once $E > 2d$.

9.2.2 The Olsen–Schaffer attractor

Takens' theorem implies we can *see* the state space of a system we are investigating: just plot the reconstructed states / delay vectors. For the SEIR model with measles parameters (Aron & Schwartz), Olsen & Schaffer (1990) reconstruct $\big(I_t, I_{t-L}, I_{t-2L}\big)$, $L = 3$ months:

9.2.3 Prediction, revisited

We observe $X_s, 0 \le s \le t$ and would like to forecast X_{t+1}. We take an approach analogous to the one we took in Section 9.1.1:

1. Build $x_s = (X_s, X_{s-1}, \ldots, X_{s-(E-1)})$ for $E - 1 \le s \le t$.
2. Find $\arg\min_{E-1 \le s \le t} \|x_s - x_t\|$.
3. Forecast X_{s+1}.

How far ahead can this work? Take two states that start out neighbors on the attractor and just let both evolve — no re-forecasting — and track how far apart they drift as a function of lead time:

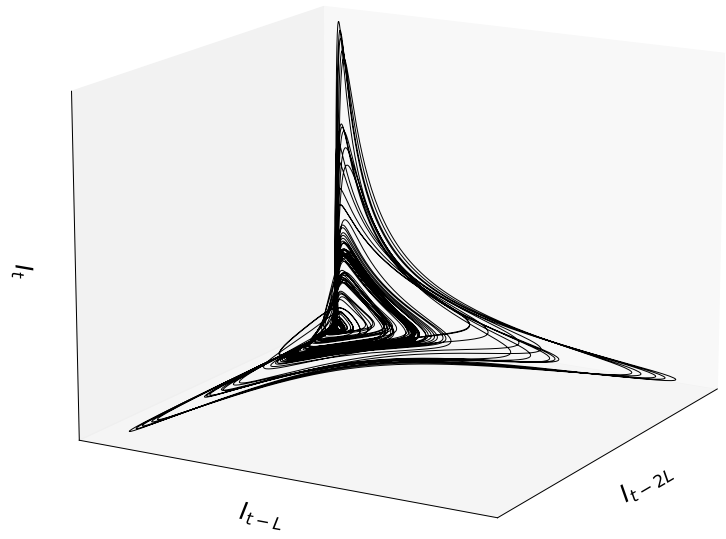


Figure 9.2: Reconstructed attractor of the seasonally forced SEIR model, measles parameters (Aron & Schwartz, as used by Olsen & Schaffer 1990): $m = 0.02\,\mathrm{yr}^{-1}$ (birth/death rate), $a = 35.84\,\mathrm{yr}^{-1}$ (latent period), $g = 100\,\mathrm{yr}^{-1}$ (infectious period), $\beta_0 = 1800\,\mathrm{yr}^{-1}$, $\beta_1 = 0.28$.

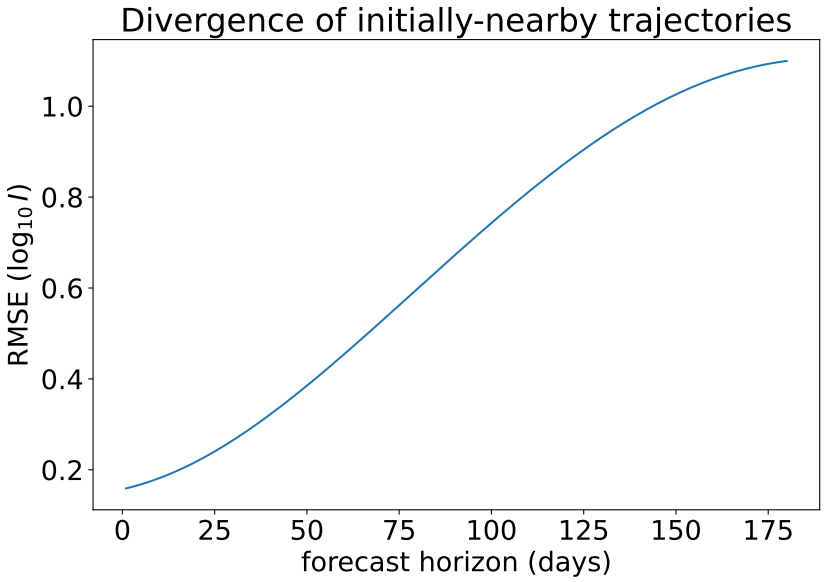


Figure 9.3: RMSE (on $\log_{10} I$) between two initially-neighboring trajectories, as a function of forecast horizon. Same reconstructed state space and parameters as Figure 9.2. Growth is monotonic: nearby states on a chaotic attractor diverge, which is exactly what bounds how far ahead *any* forecaster — nearest-neighbor or otherwise — can see.

Chapter 10

Convergent Cross Mapping

10.1 Skew products

Suppose X drives Y but not vice versa. We can write this as a set of ODEs:

$$\dot{X} = f(X), \qquad \dot{Y} = g(Y, X).$$

Or, we can write this as abstract dynamical systems:

$$\Omega^X = (\omega_1^X, \ldots, \omega_k^X) \qquad \Omega^Y = (\omega_1^X, \ldots, \omega_k^X, \omega_{k+1}^Y, \ldots, \omega_q^Y),$$

with dynamics operators T^X and T^Y. This is a *skew product*: X evolves on its own; Y's history depends on X's. We call X the *base* and Y is called the *fibre*. By Takens' theorem applied to Y, the reconstructed manifold $M_Y \cong \Omega^Y$ embeds information about X as well as information about Y. The converse reconstruction M_X does *not* contain information about Y, since X never sees Y.

10.2 Causality

Definition 14. *We say that* X causally forces Y *if* X, Y *form a skew product where* X *is the base and* Y *is the fibre.*

Operationally: can we find ω^X from ω^Y? Or, can we do the same using reconstructed states, i.e. find x_t from y_t. Importantly, this does not necessitate knowing the law or state-space of the dynamics.

10.3 Convergent Cross Mapping

Fix library length $L \leq N$. For each t in $1, \ldots, L$:

1. Find the $E+1$ nearest neighbors of $M_Y(t)$ among $\{M_Y(1), \ldots, M_Y(L)\} \setminus \{M_Y(t)\}$, with time indices $t_1, \ldots, t_{E+1}$.
2. Weight them by distance: $w_i \propto e^{-d(M_Y(t), M_Y(t_i))/d(M_Y(t), M_Y(t_1))}$.
3. Cross-map: $\hat{X}_t = \sum_i w_i X_{t_i}$.

Skill: $\rho_L = \operatorname{corr}\left(\hat{X}, X\right)$ over $t = 1, \ldots, L$.

Convergence. Plot ρ_L vs. L. If ρ_L increases and plateaus as L grows, X causally forces Y: more of Y's history fills in M_Y more completely, so nearby points on M_Y genuinely share nearby X-states. A flat or non-increasing ρ_L is evidence against causation, even if X, Y are correlated — convergence, not correlation, is the signature.

Chapter 11

Attractor Dimension

11.1 What is a dimension

Spread points uniformly across the real line and count how many fall within a distance of r from the origin. That number scales as $N(r) \sim r$.

Spread points uniformly across the plane and count how many fall within a distance of r from the origin. That number scales as $N(r) \sim r^2$.

Spread points uniformly across space and count how many fall within a distance of r from the origin. That number scales as $N(r) \sim r^3$.

For any D dimensional object, if we spread points uniformly in it and count how many are less than a distance r from any particular point —that number scales as $N(D) \sim r^D$.

Say we obeserve points in $\mathbb{R}^n$. Count their number in r balls. Then

$$N(r) \sim r^D \Rightarrow D \approx \frac{\log N(r)}{\log r}$$

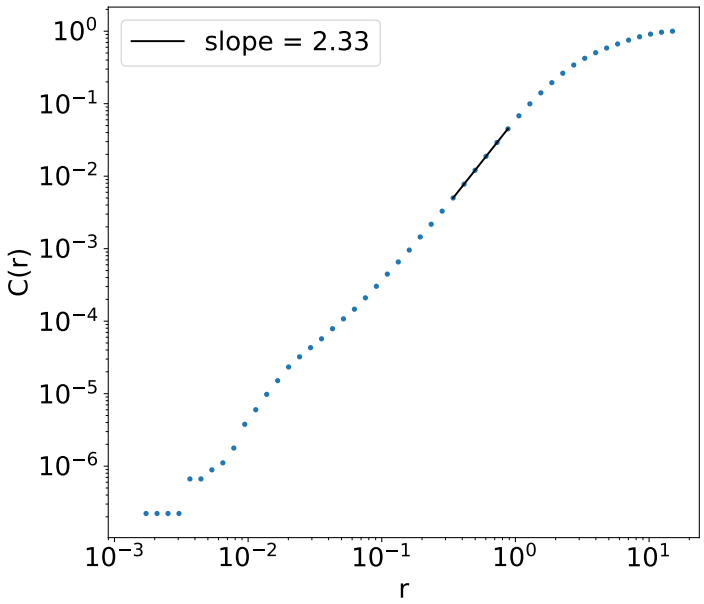
11.2 Correlation dimension

Definition 15 (Correlation sum)**.** *For points $\{x_i\}_{i=1}^N$ on the attractor,*

$$C(r) = \frac{1}{N^2} \#\{(i,j) : \|x_i - x_j\| < r\}.$$

As $r \to 0$, $C(r) \sim r^{D_C}$; the *correlation dimension* D_C is the slope of $\log C(r)$ vs. $\log r$ (Grassberger & Procaccia, 1983).

For the Olsen–Schaffer attractor (lagged $\log_{10} I$ — raw I's huge dynamic range, fine for viewing the attractor, washes out the scaling plateau here):



$D_C \approx 2.33$, close to Olsen & Schaffer's own estimate ($D_C \approx 2.52$ for this model).

11.3 Levina–Bickel MLE estimator

A second estimator, from the spacing of the k nearest neighbors. For point x_i with neighbor distances $T_1(x_i) \le \cdots \le T_k(x_i)$:

$$\hat{m}_k(x_i) = \left[\frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{T_k(x_i)}{T_j(x_i)} \right]^{-1}.$$

Average $\hat{m}_k(x_i)$ over i and over a range of k (here $k = 10, \ldots, 20$) for stability (Levina & Bickel, 2004). On the same attractor: $\hat{D}_{\mathrm{MLE}} \approx 2.48$ — consistent with D_C above.